



GOVERNMENT OF TAMILNADU

STANDARD TEN

SCIENCE

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Department of School Education

Untouchability is Inhuman and a Crime

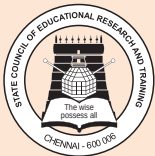
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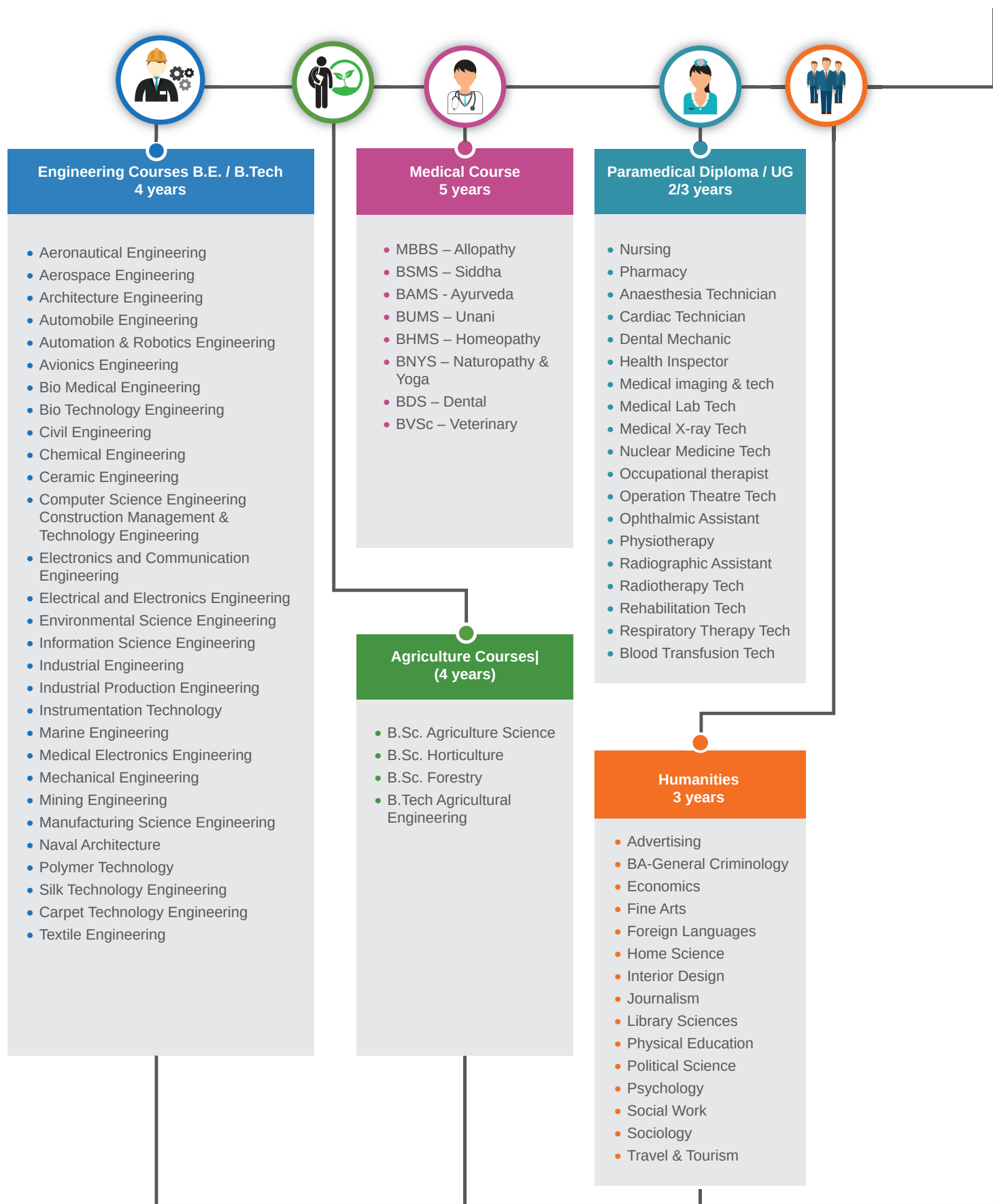
PREFACE

This book is developed in a holistic approach which inculcates comprehending and analytical skills. It will be helpful for the students to understand higher secondary science in a better way and to prepare for competitive exams in future. This textbook is designed in a learner centric way to trigger the thought process of students through activities and to make them excel in learning science.

- This book has 23 units.
- Each unit has simple activities that can be demonstrated by the teacher and also few group activities are given for students to do under the guidance of the teacher.
- Infographics and info-bits are added to enrich the learner's scientific perception.
- The "Do you know?" and "More to know" placed in the units will be an eye opener.
- Glossary has been introduced to learn scientific terms.
- ICT corner and QR code are introduced in each unit for the digital native generation.

HOW TO USE THE BOOK?

Career Guidance



→ Road ahead after 12th ...

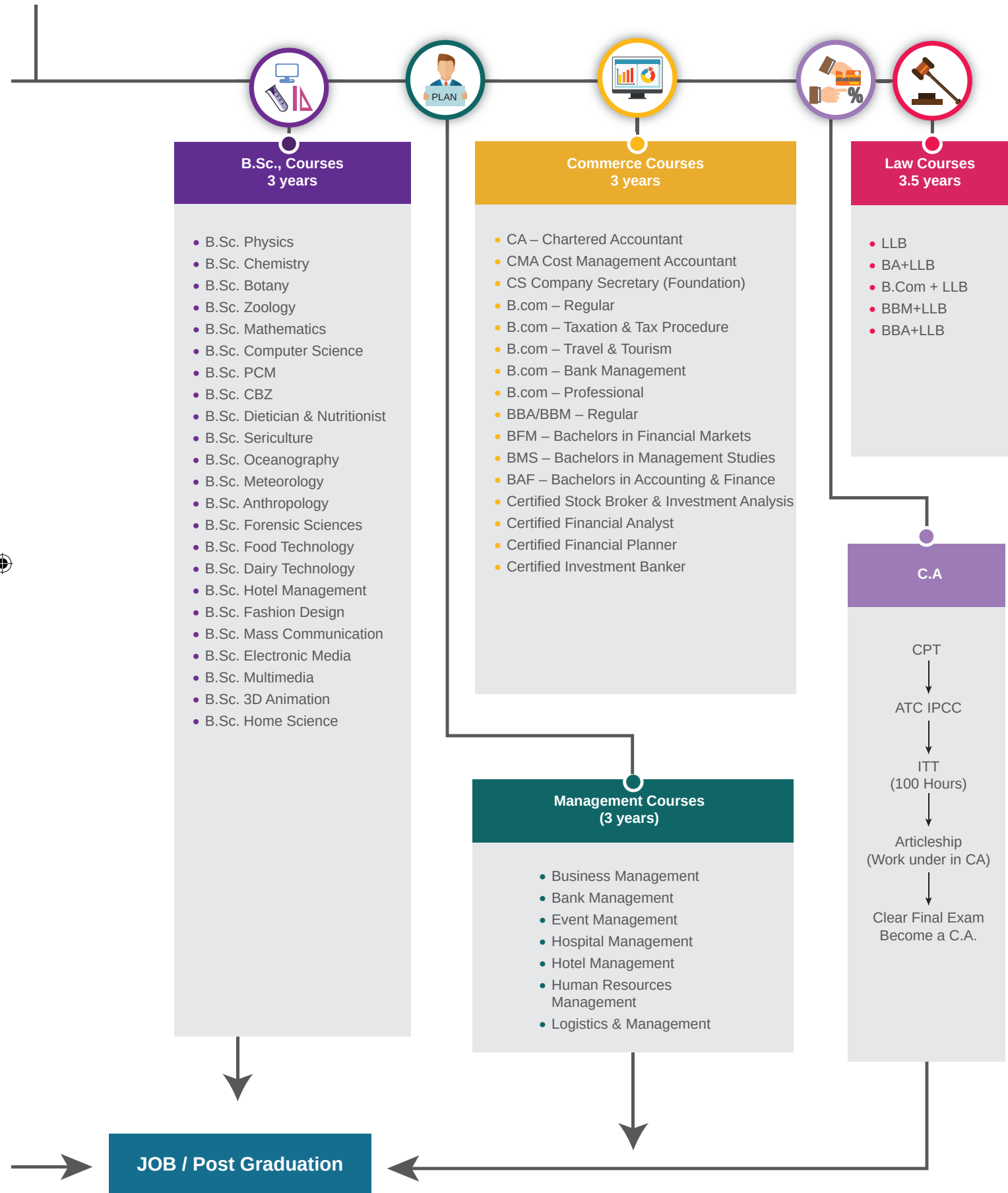


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SCIENCE



Learning Objectives



At the end of this lesson students will be able to:

- ◆ Understand the concepts of force and motion.
- ◆ Explain inertia and its types.
- ◆ State the three laws of Newton.
- ◆ Apply Newtonian concept of force and motion.
- ◆ Define force, momentum and impulse.
- ◆ Distinguish between mass and weight
- ◆ Analyze weightlessness and the principle of conservation of momentum.
- ◆ Explain the law of gravitation and its applications.
- ◆ Understand the variations in 'g' due to height and depth.
- ◆ Solve numerical problems related to force and motion

INTRODUCTION

Human beings are so curious about things around them. Things around us are related to one another. Some bodies are at rest and some are in motion. Rest and motion are interrelated terms.

In the previous classes you have learnt about various types of motion such as linear motion, circular motion, oscillatory motion, and so on. So far, you have discussed the motion of bodies in terms of their displacement, velocity, and acceleration. In this unit, let us investigate the cause of motion.

When a body is at rest, starts moving, a question that arises in our mind is 'what causes the body to move?' Similarly, when a moving object comes to rest, you would like to know what brings it to rest? If a moving object speeds

up or slows down or changes its direction. what speeds up or slows down the body? What changes the direction of motion?

One answer for all the above questions is 'Force'. In a common man's understanding of motion, a body needs a 'push' or 'pull' to move, or bring to rest or change its velocity. Hence, this 'push' or 'pull' is called as 'force'.

Let us define force in a more scientific manner using the three laws proposed by Sir Isaac Newton. These laws help you to understand the motion of a body and also to predict the future course of its motion, if you know the forces acting on it. Before Newton formulated his three laws of motion, a different perception about the force and motion of bodies prevailed. Let us first look at these ideas and then eventually learn about Newton's laws in this unit.

Mechanics is the branch of physics that deals with the effect of force on bodies. It is divided into two branches, namely, statics and dynamics.

Statics: It deals with the bodies, which are at rest under the action of forces.

Dynamics: It is the study of moving bodies under the action of forces. Dynamics is further divided as follows.

Kinematics: It deals with the motion of bodies without considering the cause of motion.

Kinetics: It deals with the motion of bodies considering the cause of motion.

1.1 FORCE AND MOTION

According to *Aristotle* a Greek Philosopher and Scientist, the natural state of earthly bodies is 'rest'. He stated that a moving body naturally comes to rest without any external influence of the force. Such motions are termed as '**natural motion**' (**Force independent**). He also proposed that a force (a push or a pull) is needed to make the bodies to move from their natural state (rest) and behave contrary to their own natural state called as '**violent motion**' (**Force dependent**). Further, he said, when two different mass bodies are dropped from a height, the heavier body falls faster than the lighter one.

Galileo proposed the following concepts about force, motion and inertia of bodies:

- The natural state of all earthly bodies is either the state of rest or the state of uniform motion.
- A body in motion will continue to be in the same state of motion as long as no external force is applied.
- When a force is applied on bodies, they resist any change in their state. This property of bodies is called 'inertia'.
- When dropped from a height in vacuum, bodies of different size, shape and mass fall at the same rate and reach the ground at the same time.

1.2 INERTIA

While you are travelling in a bus or in a car, when a sudden brake is applied, the upper part of your body leans in the forward direction. Similarly, when the vehicle suddenly is move forward from rest, you lean backward. This is due to, any body would like to continue to be in its state of rest or the state of motion. This is known as 'inertia'.

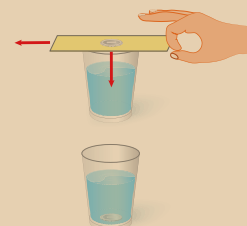
The inherent property of a body to resist any change in its state of rest or the state of uniform motion, unless it is influenced upon by an external unbalanced force, is known as 'inertia'.

Activity 1

Take a glass tumbler and place a small cardboard on it as shown in the figure. Now, keep a coin at the centre of the cardboard. Then, flick the cardboard quickly. What do you observe?

The cardboard falls off the ground and the coin falls into the glass tumbler.

Inertia of rest



In activity described above, the inertia of the coin keeps it in the state of rest when the cardboard moves. Then, when the cardboard has moved, the coin falls into the tumbler due to gravity. This happen due to 'inertia of rest'.

1.2.1 Types of Inertia

- Inertia of rest:** The resistance of a body to change its state of rest is called inertia of rest.
- Inertia of motion:** The resistance of a body to



change its state of motion is called inertia of motion.

- c) **Inertia of direction:** The resistance of a body to change its direction of motion is called inertia of direction.

1.2.2 Examples of Inertia

- ◆ An athlete runs some distance before jumping. Because, this will help him jump longer and higher. (Inertia of motion)
- ◆ When you make a sharp turn while driving a car, you tend to lean sideways, (Inertia of direction).
- ◆ When you vigorously shake the branches of a tree, some of the leaves and fruits are detached and they fall down, (Inertia of rest).

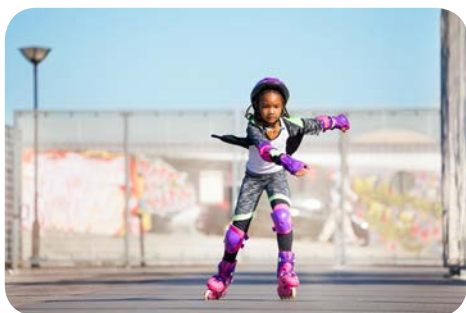


Figure 1.1 Inertia of motion

1.3 LINEAR MOMENTUM

The impact of a force is more if the velocity and the mass of the body is more. To quantify the impact of a force exactly, a new physical quantity known as linear momentum is defined. The linear momentum measures the impact of a force on a body.

The product of mass and velocity of a moving body gives the magnitude of linear momentum. It acts in the direction of the velocity of the object. Linear momentum is a vector quantity.

$$\text{Linear Momentum} = \text{mass} \times \text{velocity} \\ p = mv \dots\dots\dots (1.1)$$

It helps to measure the magnitude of a force. Unit of momentum in SI system is kg m s^{-1} and in C.G.S system its unit is g cm s^{-1} .

1.4 NEWTON'S LAWS OF MOTION

1.4.1 Newton's First Law

This law states that **every body continues to be in its state of rest or the state of uniform motion along a straight line unless it is acted upon by some external force.** It gives the definition of force as well as inertia.

1.4.2 Force

Force is an external effort in the form of push or pull, which:

1. produces or tries to produce the motion of a static body.
2. stops or tries to stop a moving body.
3. changes or tries to change the direction of motion of a moving body.

Force has both magnitude and direction. So, it is a vector quantity.

1.4.3 Types of forces

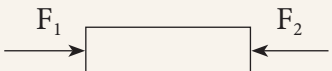
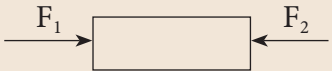
Based on the direction in which the forces act, they can be classified into two types as: (a) Like parallel forces and (b) Unlike parallel forces.

- (a) **Like parallel forces:** Two or more forces of equal or unequal magnitude acting along the same direction, parallel to each other are called like parallel forces.
- (b) **Unlike parallel forces:** If two or more equal forces or unequal forces act along opposite directions parallel to each other, then they are called unlike parallel forces. Action of forces are given in Table 1.1.

1.4.4 Resultant Force

When several forces act simultaneously on the same body, then the combined effect of the multiple forces can be represented by a single force, which is termed as '*resultant force*'. It is equal to the vector sum (adding the magnitude of the forces with their direction) of all the forces.

Table 1.1 Action of forces

Action of forces	Diagram	Resultant force (F_{net})
Parallel forces are acting in the same direction	$F_1 \longrightarrow$ $F_2 \longrightarrow$ 	$F_{\text{net}} = F_1 + F_2$
Parallel unequal forces are acting in opposite directions	$F_1 \longrightarrow$ $\longleftarrow F_2$ 	$F_{\text{net}} = F_1 - F_2$ (if $F_1 > F_2$) $F_{\text{net}} = F_2 - F_1$ (if $F_2 > F_1$) F_{net} is directed along the greater force.
Parallel equal forces are acting in opposite directions in the same line of action ($F_1 = F_2$)	$F_1 \longrightarrow$ $\longleftarrow F_2$ 	$F_{\text{net}} = F_1 - F_2$ $F_{\text{net}} = 0$ since ($F_1 = F_2$)



(a) Unlike parallel forces – Tug of war



(b) Unbalanced forces – Action of a lever



(c) Like parallel forces

Figure 1.2 Combined effect of forces

If the resultant force of all the forces acting on a body is equal to zero, then the body will be in equilibrium. Such forces are called **balanced forces**. If the resultant force is not equal to zero, then it causes the motion of the body due to **unbalanced forces**.

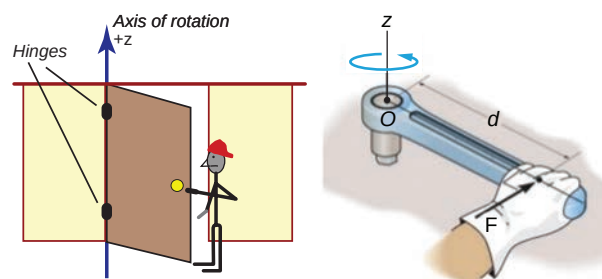
Examples: Drawing water from a well, force applied with a crow bar, forces on a weight balance, etc.

A system can be brought to equilibrium by applying another force, which is equal to the resultant force in magnitude, but opposite in direction. Such force is called as ‘**Equilibrant**’.

1.4.5 Rotating Effect of Force

Have you observed the position of the handle in a door? It is always placed at the edge of door and not at some other place. Why? Have you tried to push a door by placing your hand closer to the hinges or the fixed edge? What do you observe?

The door can be easily opened or closed when you apply the force at a point far away from the fixed edge. In this case, the effect of the force you apply is to turn the door about the fixed edge. This turning effect of the applied force is more when the distance between the fixed edge and the point of application of force is more.

**Figure 1.3** Rotating effect of a force

The axis of the fixed edge about which the door is rotated is called as the ‘**axis of rotation**’. Fix one end of a rod to the floor/wall, and apply a force at the other end tangentially.

The rod will be turned about the fixed point is called as '*point of rotation*'.

1.4.6 Moment of the Force

The rotating or turning effect of a force about a fixed point or fixed axis is called **moment of the force** about that point or **torque** (τ). It is measured by the product of the force (F) and the perpendicular distance (d) between the fixed point or the fixed axis and the line of action of the force.

$$\tau = F \times d \dots \dots \dots (1.2)$$

Torque is a vector quantity. It is acting along the direction, perpendicular to the plane containing the line of action of force and the distance. Its SI unit is Nm.

Couple: Two equal and unlike parallel forces applied simultaneously at two distinct points constitute a couple. The line of action of the two forces does not coincide. It does not produce any translatory motion since the resultant is zero. But, a couple results in causes the rotation of the body. Rotating effect of a couple is known as **moment of a couple**.

Examples: Turning a tap, winding or unwinding a screw, spinning of a top, etc.

Moment of a couple is measured by the product of any one of the forces and the perpendicular distance between the line of action of two forces. The turning effect of a couple is measured by the magnitude of its moment.

Moment of a couple = Force $\times$ perpendicular distance between the line of action of forces

$$M = F \times S \dots \dots \dots (1.3)$$

The unit of moment of a couple is newton metre (N m) in SI system and dyne cm in CGS system.

By convention, the direction of moment of a force or couple is taken as positive if the body is rotated in the anti-clockwise direction and

negative if it is rotate in the clockwise direction. They are shown in Figures 1.4 (a and b)

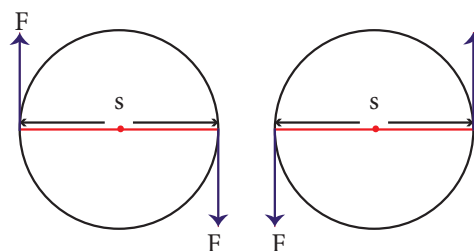


Figure 1.4 (a)

Figure 1.4 (b)

Clockwise moment Anticlockwise moment

1.4.7 Application of Torque

1. Gears:

A gear is a circular wheel with teeth around its rim. It helps to change the speed of rotation of a wheel by changing the torque and helps to transmit power.



2. Seasaw

Most of you have played on the seasaw. Since there is a difference in the weight of the persons sitting on it, the heavier person lifts the lighter person. When the heavier person comes closer to the pivot point (fulcrum) the distance of the line of action of the force decreases. It causes less amount of torque to act on it. This enables the lighter person to lift the heavier person.

3. Steering Wheel

A small steering wheel enables you to manoeuvre a car easily by transferring a torque to the wheels with less effort.

1.4.8 Principle of Moments

When a number of like or unlike parallel forces act on a rigid body and the body is in equilibrium, then the algebraic sum of the moments in the clockwise direction is equal to the algebraic sum of the moments in the anticlockwise direction. In other words, at

equilibrium, the algebraic sum of the moments of all the individual forces about any point is equal to zero.

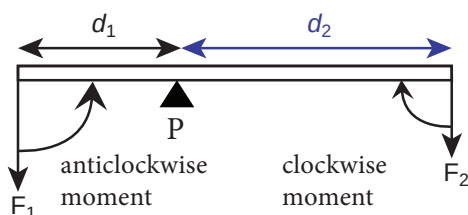


Figure 1.5 Principle of moments

In the illustration given in figure 1.5, the force F_1 produces an anticlockwise rotation at a distance d_1 from the point of pivot P (called fulcrum) and the force F_2 produces a clockwise rotation at a distance d_2 from the point of pivot P. The principle of moments can be written as follows:

Moment in clockwise direction = Moment in anticlockwise direction

$$F_1 \times d_1 = F_2 \times d_2 \dots\dots\dots (1.4)$$

1.5 NEWTON'S SECOND LAW OF MOTION

According to this law, “**the force acting on a body is directly proportional to the rate of change of linear momentum of the body and the change in momentum takes place in the direction of the force**”.

This law helps us to measure the amount of force. So, it is also called as ‘*law of force*’. Let, ‘ m ’ be the mass of a moving body, moving along a straight line with an initial speed ‘ u ’. After a time interval of ‘ t ’, the velocity of the body changes to ‘ v ’ due to the impact of an unbalanced external force F .

Initial momentum of the body $P_i = mu$

Final momentum of the body $P_f = mv$

Change in momentum $\Delta p = P_f - P_i$
 $= mv - mu$

By Newton's second law of motion,

Force, $F \propto$ rate of change of momentum

$F \propto$ change in momentum / time

$$F \propto \frac{mv - mu}{t}$$

$$F = \frac{km(v - u)}{t}$$

Here, k is the proportionality constant.
 $k = 1$ in all systems of units. Hence,

$$F = \frac{m(v - u)}{t} \quad (1.5)$$

Since, acceleration = change in velocity / time, $a = (v - u) / t$. Hence, we have

$$F = m \times a \quad (1.6)$$

Force = mass $\times$ acceleration

No external force is required to maintain the motion of a body moving with uniform velocity. When the net force acting on a body is not equal to zero, then definitely the velocity of the body will change. Thus, change in momentum takes place in the direction of the force. The change may take place either in magnitude or in direction or in both.

Force is required to produce the acceleration of a body. In a uniform circular motion, even though the speed (magnitude of velocity) remains constant, the direction of the velocity changes at every point on the circular path. So, the acceleration is produced along the radius called as *centripetal acceleration*. The force, which produces this acceleration is called as centripetal force, about which you have learnt in class IX.

Units of force: SI unit of force is newton (N) and in C.G.S system its unit is dyne.

Definition of 1 newton (N): The amount of force required for a body of mass 1 kg produces an acceleration of 1 m s^{-2} , **1 N = 1 kgms⁻²**

Definition of 1 dyne: The amount of force required for a body of mass 1 gram produces an acceleration of 1 cm s^{-2} , **1 dyne = 1 gcms⁻²**; also **1 N = 10⁵ dyne**.

Unit force:

The amount of force required to produce an acceleration of 1 ms^{-2} in a body of mass 1 kg is called 'unit force'.

Gravitational unit of force:

In the SI system of units, gravitational unit of force is kilogram force, represented by kg f . In the CGS system its unit is gram force, represented by g f .

$$1 \text{ kgf} = 1 \text{ kg} \times 9.8 \text{ ms}^{-2} = 9.8 \text{ N};$$

$$1 \text{ gf} = 1 \text{ g} \times 980 \text{ cms}^{-2} = 980 \text{ dyne}$$

1.6 Impulse

A large force acting for a very short interval of time is called as 'Impulsive force'. When a force F acts on a body for a period of time t , then the product of force and time is known as 'impulse' represented by ' J '

$$\text{Impulse, } J = F \times t \quad (1.7)$$

By Newton's second law

$$F = \Delta p / t \quad (\Delta \text{ refers to change})$$

$$\Delta p = F \times t \quad (1.8)$$

From 1.7 and 1.8

$$J = \Delta p$$

Impulse is also equal to the magnitude of change in momentum. Its unit is kgms^{-1} or Ns .

Change in momentum can be achieved in two ways. They are:

- a large force acting for a short period of time and
- a smaller force acting for a longer period of time.

Examples:

- ◆ Automobiles are fitted with springs and shock absorbers to reduce jerks while moving on uneven roads.
- ◆ In cricket, a fielder pulls back his hands while catching the ball. He experiences a smaller force for a longer interval of time to catch the ball, resulting in a lesser impulse on his hands.



Figure 1.6 Example of impulsive force

1.7 NEWTON'S THIRD LAW OF MOTION

Newton's third law states that '**for every action, there is an equal and opposite reaction. They always act on two different bodies**'.

If a body A applies a force F_A on a body B, then the body B reacts with force F_B on the body A, which is equal to F_A in magnitude, but opposite in direction. $F_B = -F_A$

Examples:

- ◆ When birds fly they push the air downwards with their wings (Action) and the air pushes the bird upwards (Reaction).
- ◆ When a person swims he pushes the water using the hands backwards (Action), and the water pushes the swimmer in the forward direction (Reaction).
- ◆ When you fire a bullet, the gun recoils backward and the bullet is moving forward (Action) and the gun equalises this forward action by moving backward (Reaction).

1.8 PRINCIPLE OF CONSERVATION OF LINEAR MOMENTUM

There is no change in the linear momentum of a system of bodies as long as no net external force acts on them.

